# Translations

Exam code: 0607

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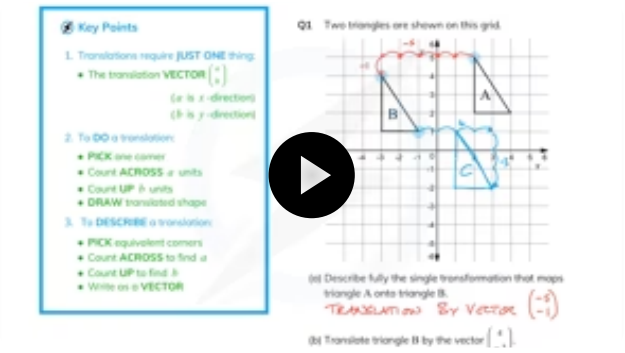


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## Translations

### What are transformations in maths?

- There are **four transformations** to learn
  - **translations**, **rotations**, **reflections** and **enlargements**
- A transformation can **change** the **position**, **orientation** and/or **size** of a **shape**
  - The **original shape** is called the **object**
  - The **transformed shape** is called the **image**
- Vertices are labelled to show **corresponding points**
  - Vertices on the **object** are labelled A, B, C, etc.
  - Vertices on the **image** are labelled A', B', C' etc.

### What is a translation?

- A **translation moves** a shape
- The **size** and **orientation** (which way up it is) of the shape **stays the same**
  - The **object** and **image** are **congruent**

### What is a translation vector?

- The movement of a translation is described by a **vector**
- You need to know how to write a translation using a vector (rather than words)

**Stuck on something?**

Ask questions and get help with exactly what you're revising.

Got it

- $x$ is the distance moved **horizontally**
  - **Negative** means move to the **left**
  - **Positive** means move to the **right**
- $y$ is the distance moved **vertically**
  - **Negative** means move **down**
  - **Positive** means move **up**

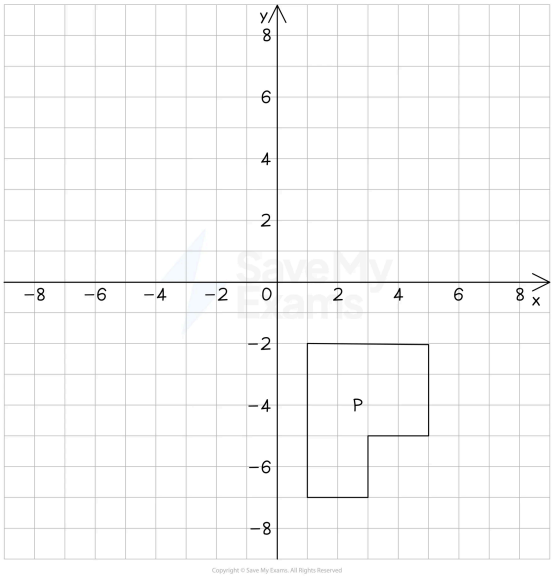
## How do I translate a shape?

- **STEP 1**
  **Interpret** the translation vector
  - $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$ means 3 to the **right** and 1 **down**
- **STEP 2**
  **Move each vertex** on the **original object** according to the vector
- **STEP 3**
  **Connect** the **new vertices** and label the **translated image**
  - It should look **identical** to the **original object** just in a **different position**
- In some cases the **image** can overlap the **object**

### Worked Example

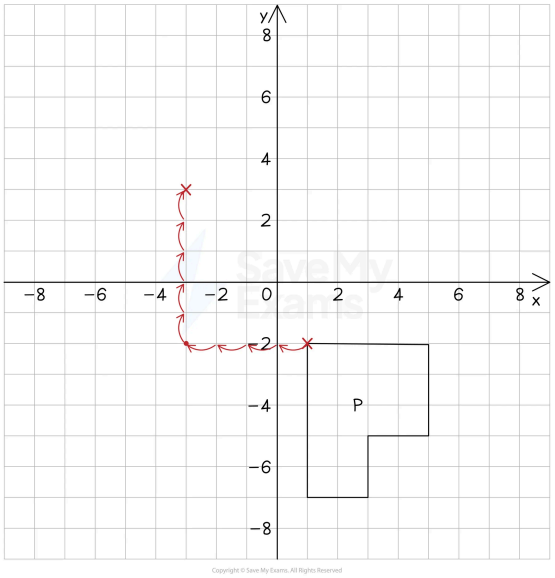
On the grid below translate shape P using the vector $\begin{pmatrix} -4 \\ 5 \end{pmatrix}$.

Label your translated shape P'.

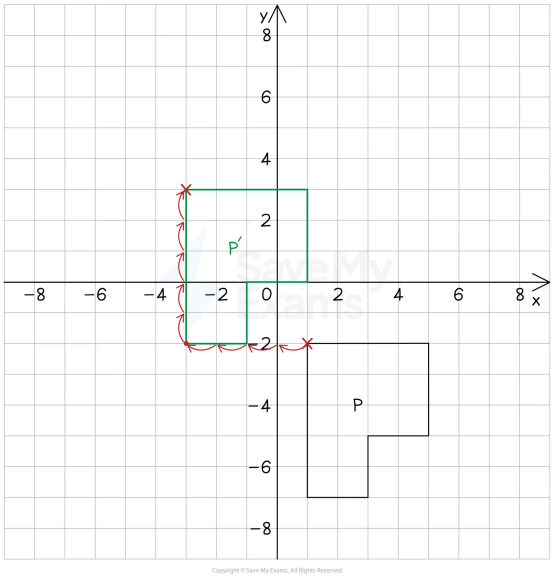



**Answer:**

The vector means "4 to the left" and "5 up"
You don't have to draw in any arrows but it is a good idea to




Translating one vertex and then following around the shape one vertex at a time makes it easier to get the shape in exactly the right position!




## How do I describe a translation?

- To describe a **translation**, you must:
  - State that the transformation is a **translation**
  - Give the **column vector** that describes the movement
- To find the **vector**:
  - **Pick** a **point** on the **original** shape
  - **Identify** the **corresponding point** on the **image**
  - Count how far **left or right** ($x$) you need to go **from the object** to get **to the image**
    - If you go to the **left** then $x$ will be a **negative number**

- Write these numbers as a vector
  - $\begin{pmatrix} x \\ y \end{pmatrix}$

### Examiner Tips and Tricks

The vector is how the shape moves, **not** the size of the gap between the object and the image. Watch out for this common error!

Use **tracing paper** to check your answer.

### Worked Example

Describe fully the single transformation that creates shape B from shape A.

**Answer:**

This is a case where the image overlaps the object
You should still see that the shape is the same size and the same way up so it is a **translation**
Start at a vertex on the **original object** that is well away from any overlap area to avoid confusion and count the number of position left/right and up/down that you need to move to reach the **corresponding vertex** on the translated image
Take care when counting around the axes!

**Shape A has been translated using the vector** $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$

## How do I reverse a translation?

- To **return a shape to its original position** after a translation
  - the **horizontal** and **vertical** translations must **both** be **reversed**
- The **column vector** to **reverse a translation** is simply the same as the original vector, but with the **sign of both values changed**
  - E.g. For a translation described by the column vector $\begin{pmatrix} -2 \\ 7 \end{pmatrix}$
  - The column vector for the reverse translation is $\begin{pmatrix} 2 \\ -7 \end{pmatrix}$

### Worked Example

A shape has been translated from A to B using the translation vector $\begin{pmatrix} 6 \\ -8 \end{pmatrix}$.

**Answer:**

Write down the vector used to reverse the translation from B to A.

To reverse the translation the shape must be moved in the opposite direction by the same amount

Reverse the signs in the translation vector

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